COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 12

Combinational Circuit

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MUX

Question Discussion



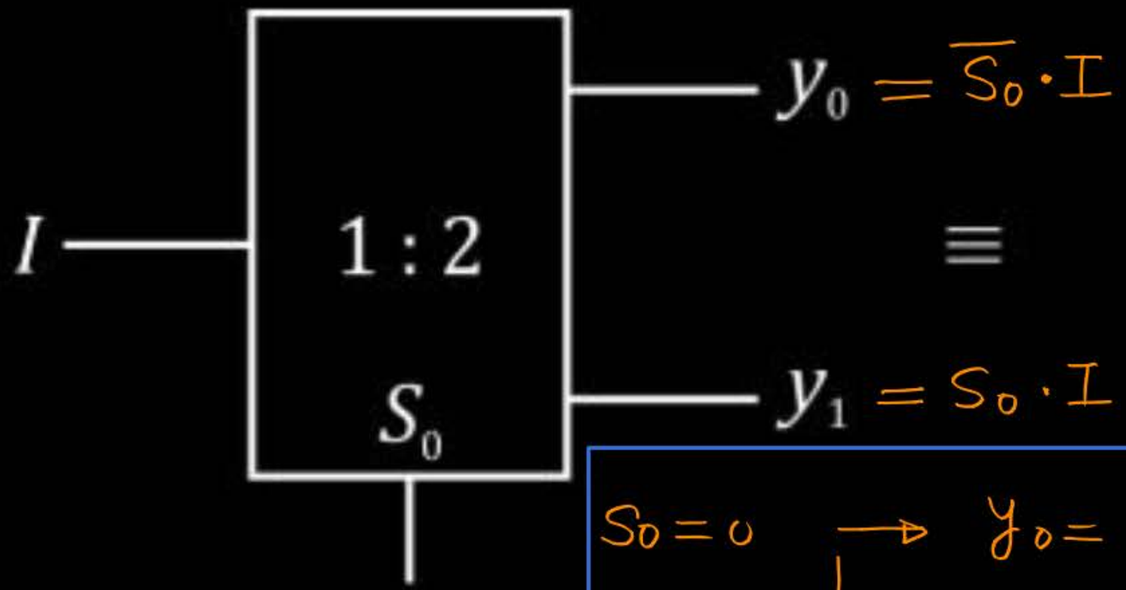
- DeMUX
- Decoder

DeMUX

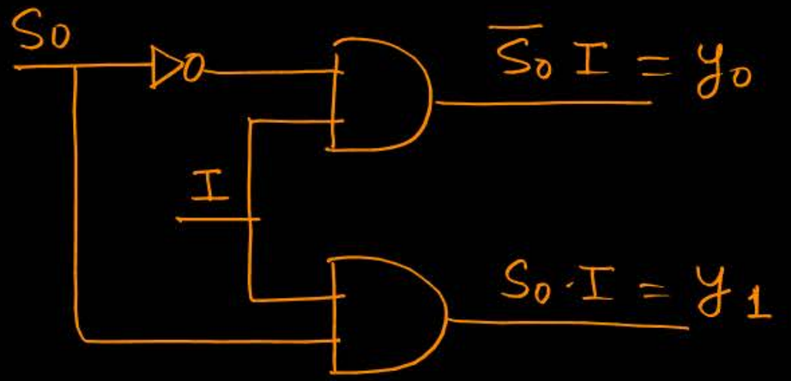
What is DeMUX ?

It is combinational CKt having one-i/p and many o/p lines. And on the basis of select i/p lines i/p line is transferred to one of the o/p line

- 1 : 2 DeMUX



Internal Circuit



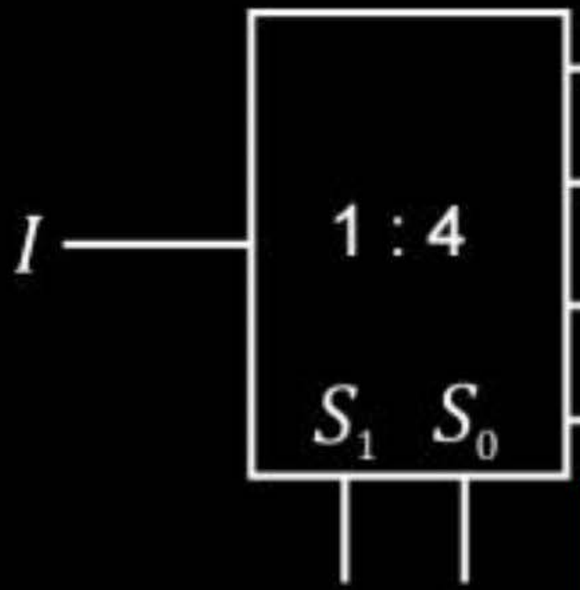
$S_0 = 0 \longrightarrow y_0 = 1 \cdot I = I$

$y_1 = 0 \cdot I = 0$

$S_0 = 1 \longrightarrow y_0 = 0 \cdot I = 0$

$y_1 = 1 \cdot I = I$

- 1 : 4 DeMUX



$y_0 = \overline{S_1} \cdot \overline{S_0} \cdot I$

$y_1 = \overline{S_1} S_0 \cdot I$

$\equiv$

$y_2 = S_1 \overline{S_0} \cdot I$

$y_3 = S_1 S_0 \cdot I$

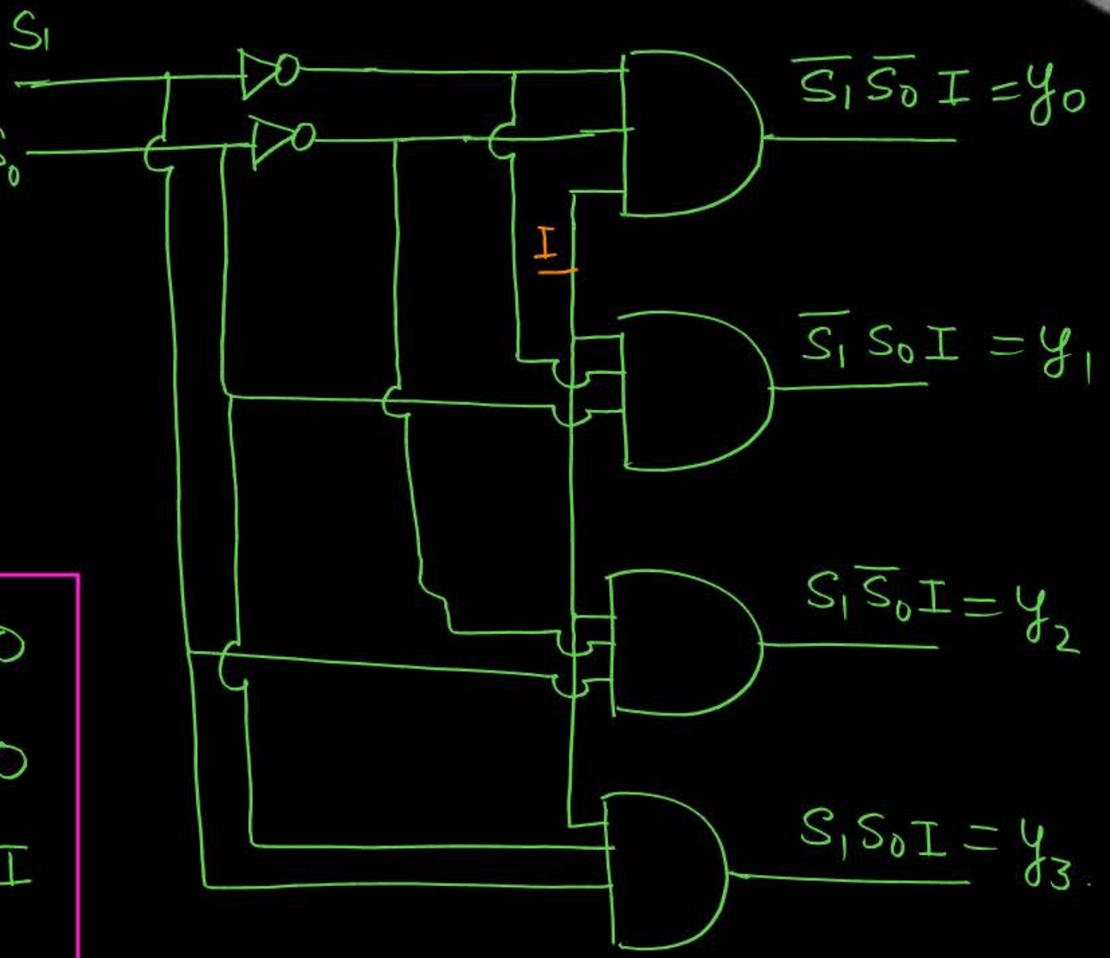
$S_1 = 1, S_0 = 0, \quad y_0 = 0$

$y_1 = 0$

$y_2 = I$

$y_3 = 0$

Internal Circuit



Higher order DeMUX using lower order DeMUX

- $1:4$ $1:2^2$ $\xrightarrow{\text{Using } 1:2}$ $2+1 = (2^2-1)$
- $1:8$ $1:2^3$ $\xrightarrow{\text{Using } 1:2}$ $4+2+1 = (2^3-1)$
- $1:16$ $1:2^4$ $\xrightarrow{\text{Using } 1:2}$ $8+4+2+1 = (2^4-1)$
- $1:2^n$ $\xrightarrow{\text{Using } 1:2}$ (2^n-1)

- 1 : 16 $\xrightarrow{\text{Using } 1:4}$ $4 + 1$
- 1 : 32 $\xrightarrow{\text{Using } 1:4}$ $(8 + 2)$

  $10\,(1:4) \;+\; 1\,(1:2)$
- 1 : 64 $\xrightarrow{\text{Using } 1:8}$ $8 + 1$
- 1 : 256 $\xrightarrow{\text{Using } 1:8}$ $(32 + 4)$

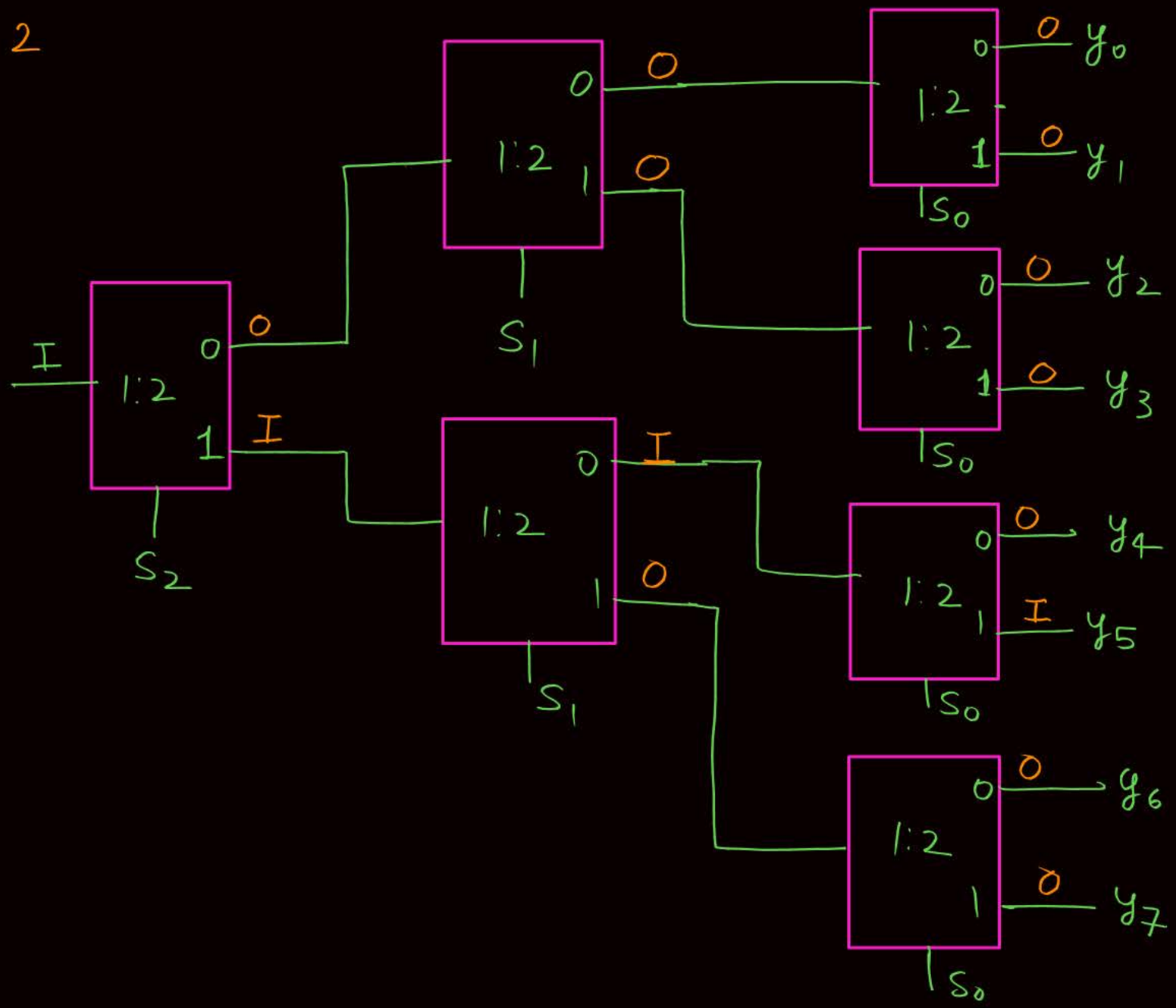
  $36\,(1:8) + 1\,(1:4)$

1:8 DeMUX using 1:2

I, S_2, S_1, S_0

1:8 $\xrightarrow{1:2}$ 4 + 2 + 1

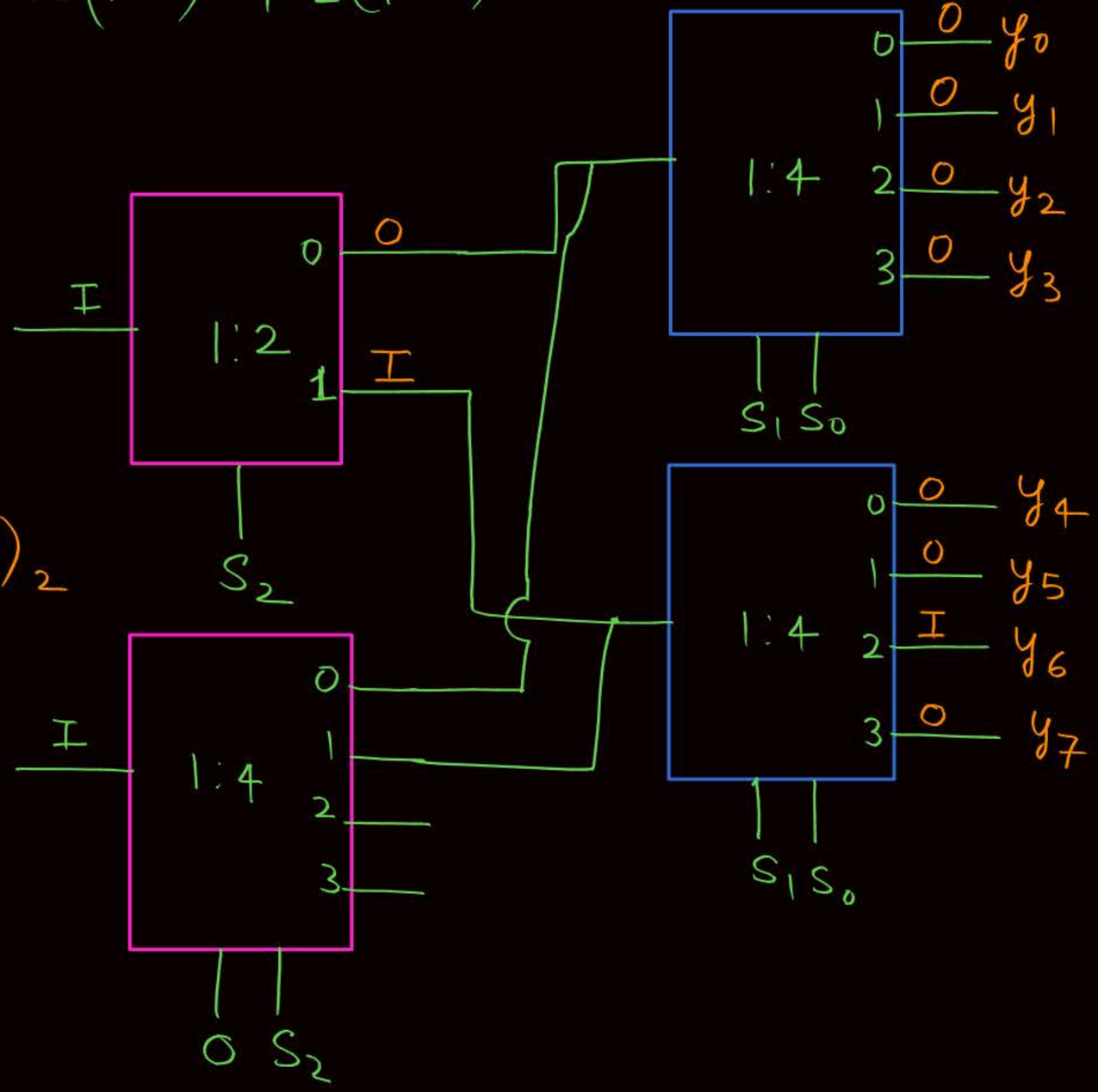
$S_2 S_1 S_0 = (101)_2$



1:8 $\xrightarrow{1:4}$ 2(1:4) + 1(1:2)

I, S_2, S_1, S_0

$S_2 S_1 S_0 = (110)_2$



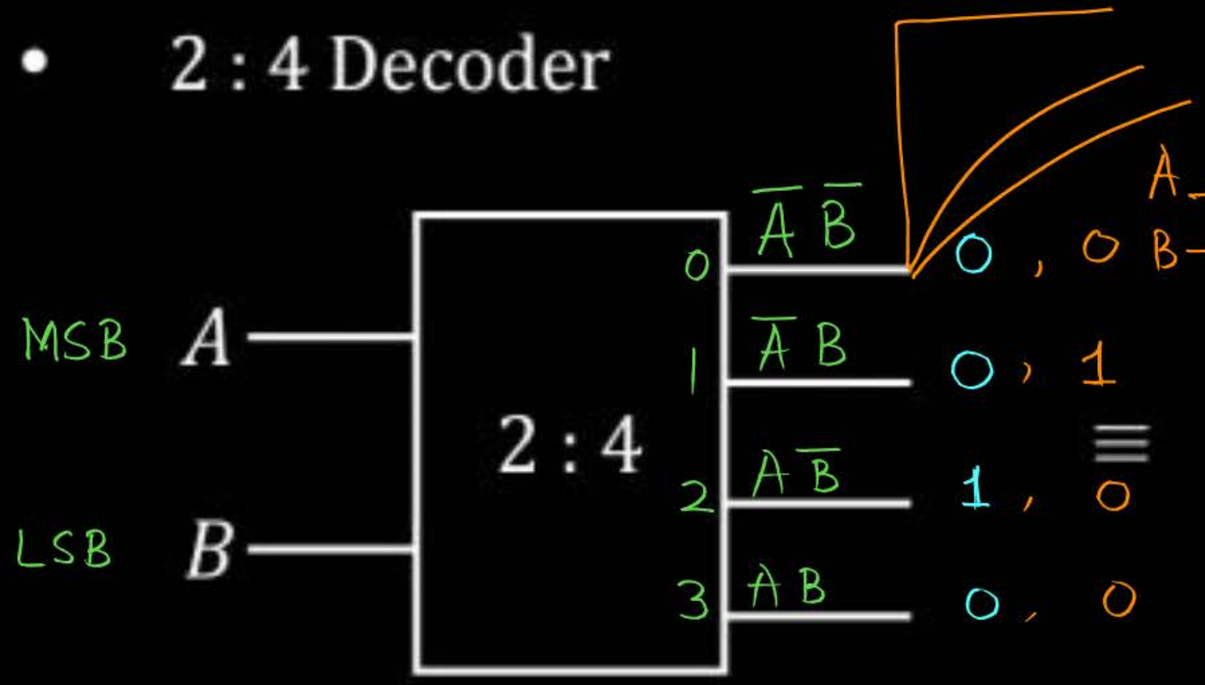
Decoder

- What is Decoder ?

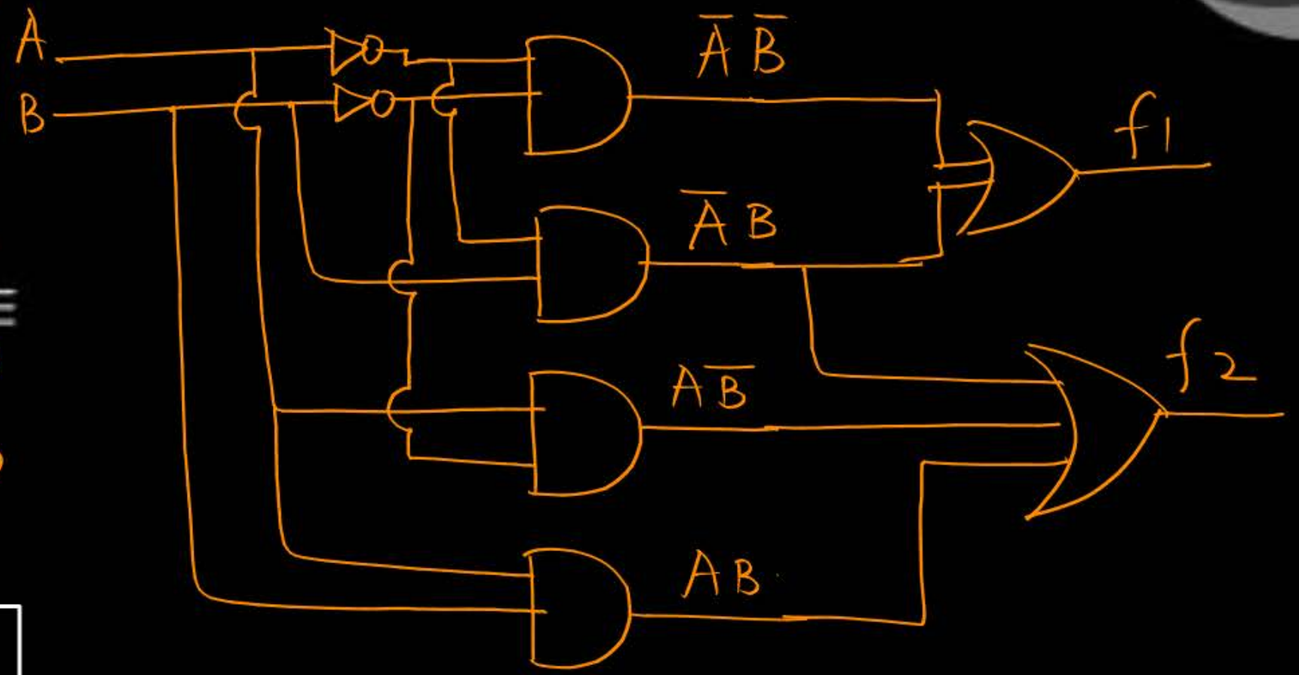
- It is combinational ckt having many i/Ps and many o/Ps. It is basically used to convert binary into other codes.
  - 3:8 decoder → binary to Octal
  - 4:16 decoder → binary to hexadecimal.

Decoder [order] $(n : 2^n)$

- 2 : 4 Decoder



Internal Circuit



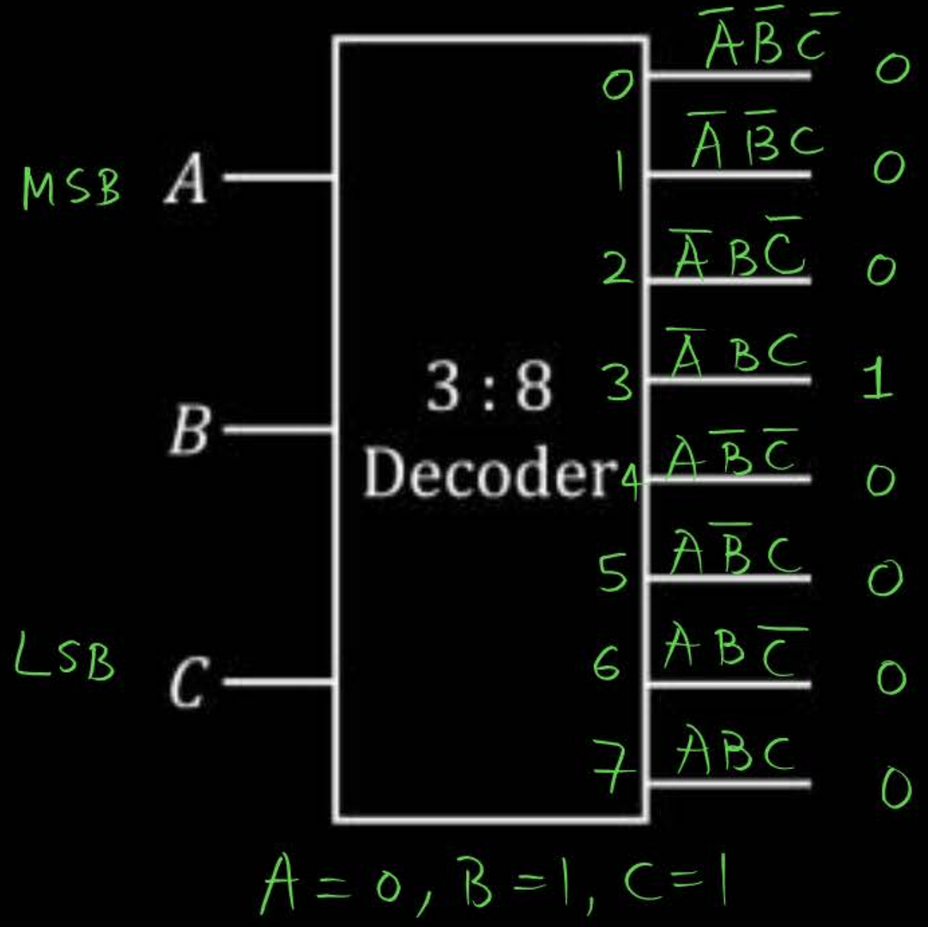
$A = 1$, $B = 0$ ⟶ PIN 2 → is 1 only

$A = 0$, $B = 1$ ⟶ PIN 1 → is 1 only

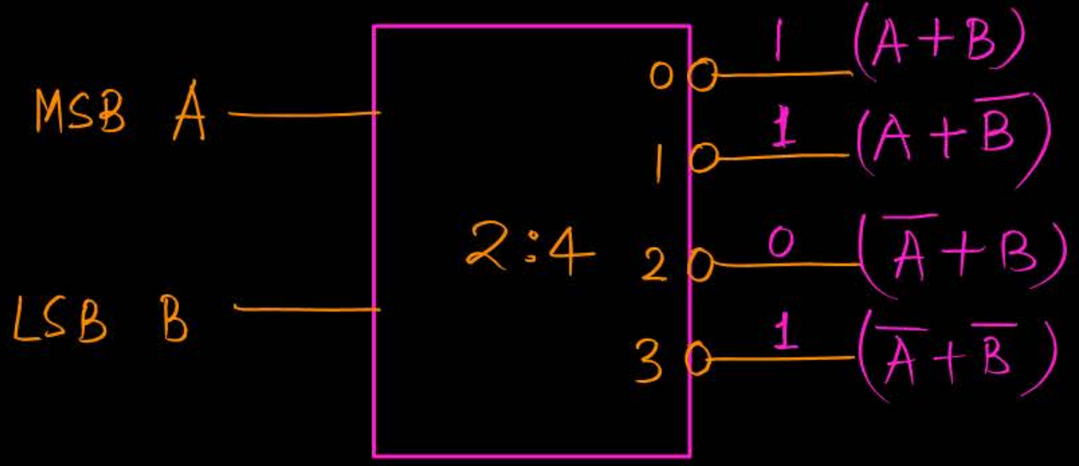
$f_1(A,B) = \Sigma(0,1) = \bar{A}\bar{B} + \bar{A}B$

$f_2(A,B) = \Sigma(1,2,3) = \bar{A}B + A\bar{B} + AB$

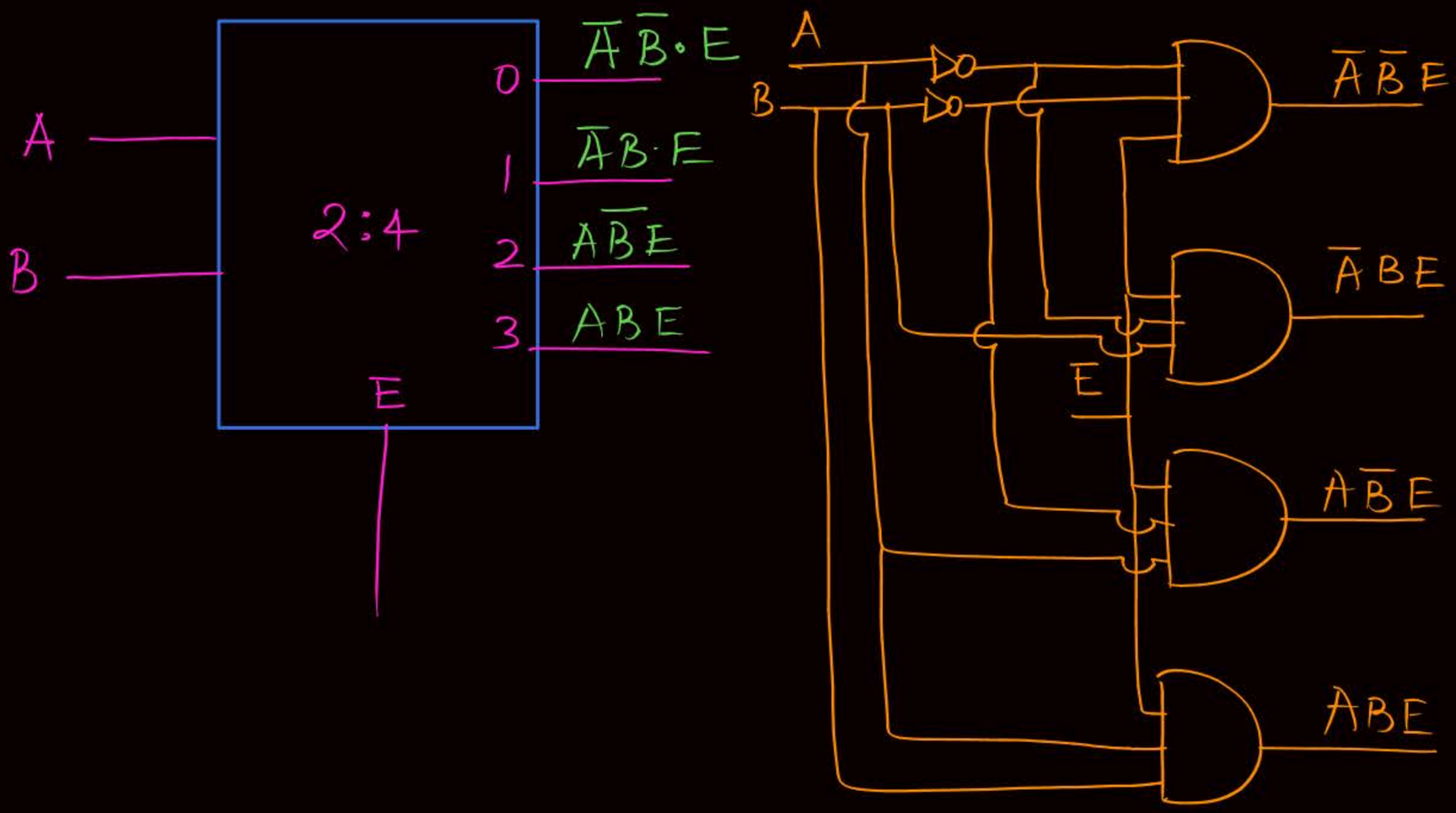
- 3 : 8 Decoder



$A = 0, B = 1, C = 1$



A=1, B=0 ⟶ PIN 2 will be '0' only.


A
B
2:4
E
0
1
2
3
$\bar{A}\bar{B}\cdot E$
$\bar{A}B\cdot E$
$A\bar{B}E$
ABE
A
B
E
$\bar{A}\bar{B}E$
$\bar{A}BE$
$A\bar{B}E$
ABE

Comparison of DeMUX and Decoder circuit

- How their internal circuits are same ?

- Decoder with enable input pin have same internal circuitery as of DeMUX of equivalent order.

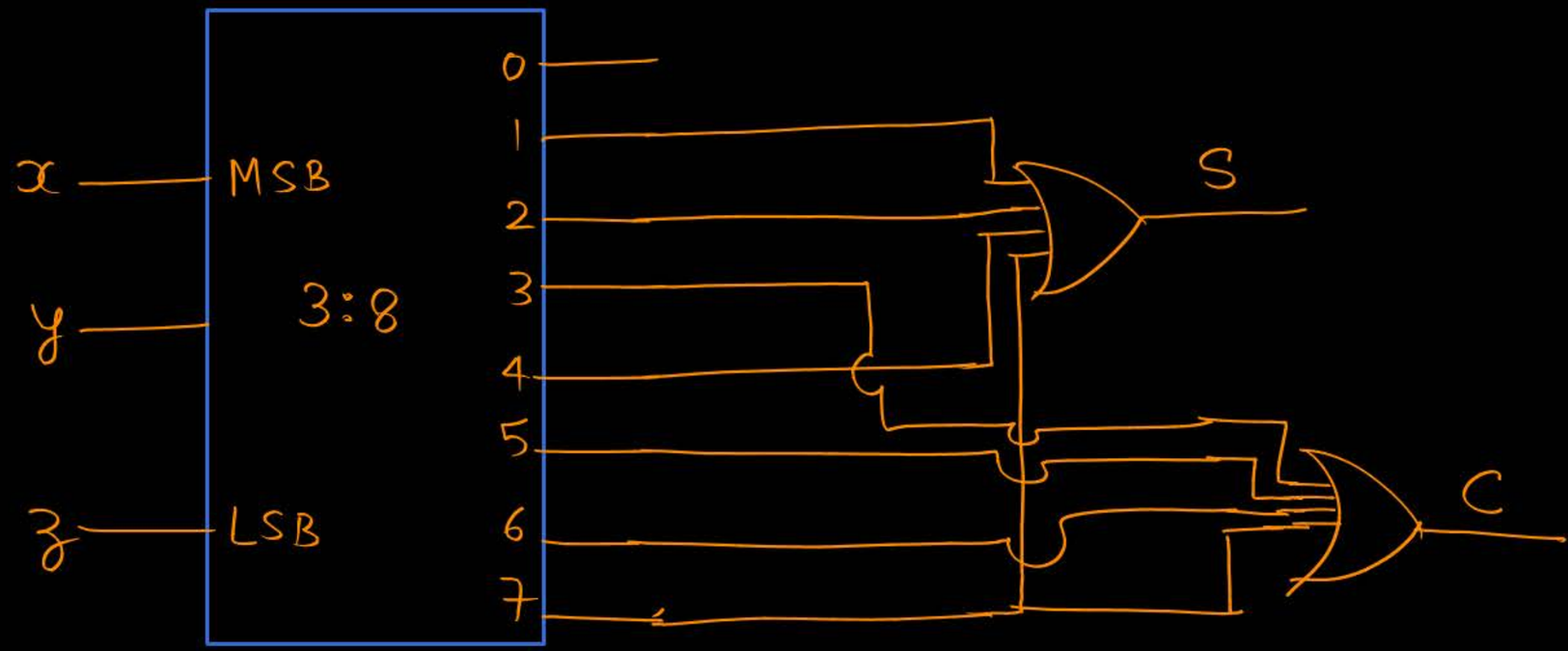
Full adder implementation using Decoder + OR gate

- Full adder using 3 : 8 decoder + OR gate :

(x, y, z)

$S(x, y, z) = \Sigma(1, 2, 4, 7) = x \oplus y \oplus z$

$C(x, y, z) = \Sigma(3, 5, 6, 7) = xy + yz + zx = xy + (x \oplus y)z$



- $f(A,\bar{B},C) = AB+BC+CA = \Sigma(3,5,6,7)$

$f(\bar{A},\bar{B},\bar{C}) = f_1(A,B,C) = \overline{f(A,B,C)} = \overline{AB+BC+AC}$

$= \Sigma(0,1,2,4)$

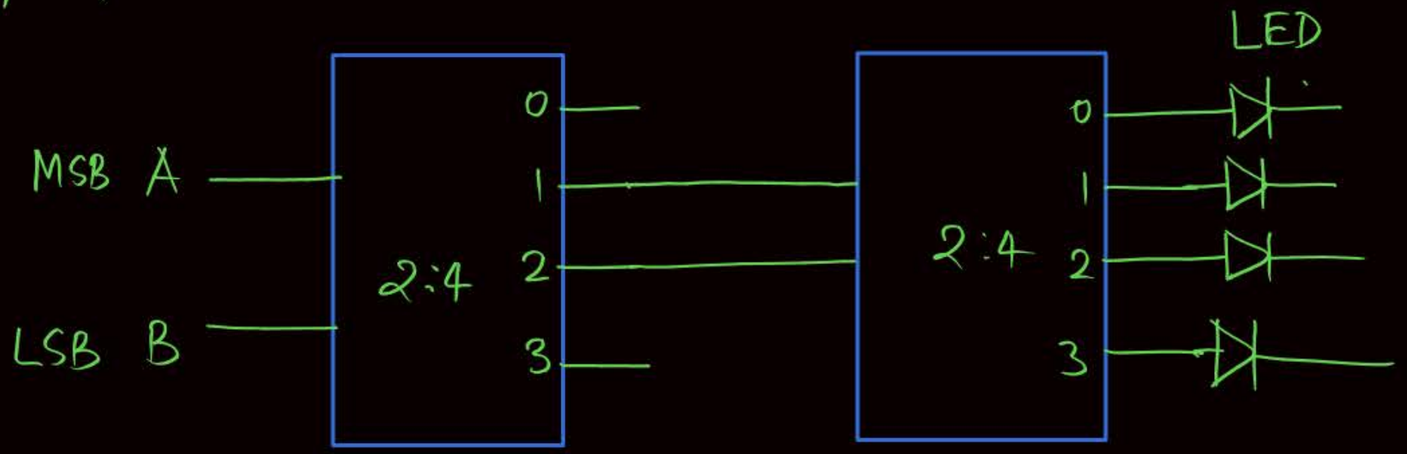
$= (\bar{A}+\bar{B})(\bar{B}+\bar{C})(\bar{A}+\bar{C})$

$= (P+Q)(Q+R)(P+R)$

$= \bar{A}\bar{B}+\bar{B}\bar{C}+\bar{A}\bar{C}$

$= PQ+QR+PR$

#Q.



On i/p side we apply the different combination of A, B repeatedly then which no. o/p LED will never glow ____.



2 Minute Summary

- DeMUX
- Decoder

Thank you

GW

Soldiers !

